

GATE

CRASH COURSE

ALL BRANCHES

**Engineering
Mathematics**

Lecture 07

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Topics *to be covered*

PROBABILITY DISTRIBUTIONS



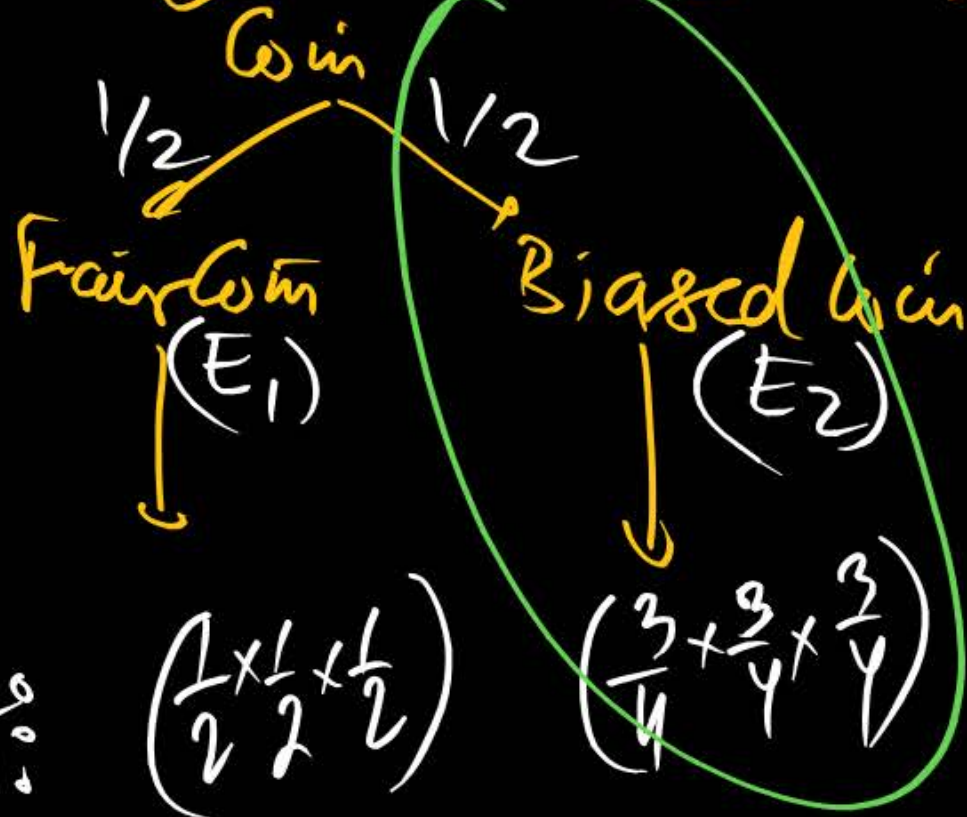
[NAT]

HWD

Mahesh has two coins, one is fair and the other is biased. The fair coin has a 50% chance of head, while the biased coin has a 75% chance of heads. One of the coins is randomly chosen and flipped 3 times. All 3 flips result in heads. What is the probability that the coin is biased?

HW

- (a) 0.421 (b) 0.125 (c) 0.771 (d) 0.273



All three tosses are (HHH) Ans

Fair Coin $\begin{cases} P(H) = \frac{1}{2} \\ P(T) = \frac{1}{2} \end{cases}$

Biased Coin $\begin{cases} P(H) = \frac{3}{4} \\ P(T) = \frac{1}{4} \end{cases}$

$$P(E_2/A) = \frac{\frac{1}{2} \times \left(\frac{3}{4}\right)^3}{\frac{1}{2} \times \left(\frac{1}{2}\right)^3 + \frac{1}{2} \times \left(\frac{3}{4}\right)^3}$$
$$= 0.771 \quad (c) \checkmark$$

Statistics

Random Variable → whenever we are not sure about the outcome of an Experiment, then such types of Experiments are called R-Exp & Variable involve in R-Exp is called RANDOM VARIABLE.

Discrete R.V (x) → Counting Related Variables are called D.R.V.

for eg Number of students, No. of vehicles, No. of Deaths etc

Continuous Random Variable (x) → when R.V has infinite possibilities in a certain Range then it is called C.R.V

for eg, Height, weight, Age, time etc

Discrete R.V.

$$\begin{array}{l}
 X : \begin{bmatrix} x_1 & x_2 & x_3 & \dots & x_n \\ p_1 & p_2 & p_3 & \dots & p_n \end{bmatrix} \\
 P(X) :
 \end{array}$$

$$E(X) = \text{Mean}(X) = \sum p_i x_i = \frac{\sum x_i}{n}$$

$$2^{\text{nd}} \text{ Moment} = E(X^2) = \sum p_i x_i^2$$

$$\text{Var}(X) = E(X^2) - E^2(X)$$

$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

$$\textcircled{*} \sum p_i = 1, p_i \geq 0$$

$$\textcircled{*} \text{Md} = ? = \left(\frac{n+1}{2}\right)^{\text{th}} \text{ for } n = \text{odd}$$

$$= \left(\frac{n}{2}\right)^{\text{th}} + \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ for } n = \text{even}$$

$$\textcircled{*} \text{Var}(X) \geq 0$$

$$\text{or } \text{SD} \geq 0$$

$$\textcircled{*} \text{Var}(ax+by) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X,Y)$$

$$\textcircled{*} \text{Cov}(X,Y) = E(XY) - E(X) \cdot E(Y)$$

Mode: Data having Highest Frequency is called Mode

The mean of the observations of the first 50 observations of a process is 12. If the 51st observation is 18 then, the mean of the first 51 observations of the process is

(A) 12

$$\bar{X}_{51} = \frac{\sum X_{50} + 18}{51}$$

(B) 12.12

(C) 12.36

$$= \frac{(\bar{X}_{50} \times 50) + 18}{51}$$

(D) 18

$$= \frac{12 \times 50 + 18}{51} = 12.11$$

(b) ✓

[MSQ]

Let A and B are two independent random variables.
Which of the following statements is/are incorrect?

- (a) $E(A+B)^2 = E(A)^2 + E(B)^2 + 2E(A) \cdot E(B)$ *True*
- (b) $\text{Var}(A+B) = \text{Var}(A) + \text{Var}(B) + 2E(A) \cdot E(B)$ *false*
- (c) $E(A^2 + B^2) = \text{Var}(A) + \text{Var}(B) + E(A)^2 + E(B)^2$ *True*
- (d) $E(A+B)^2 = E(A^2) + E(B^2)$ *false*

$$(a) E(A+B)^2 = E(A^2 + B^2 + 2AB) = E(A^2) + E(B^2) + 2E(A) \cdot E(B) \text{ True}$$

$$(b) \text{Var}(A+B) = \text{Var}(A) + \text{Var}(B) + 2\text{Cov}(A, B)$$

$$(c) E(A^2) + E(B^2) = (E(A^2) - E^2(A)) + (E(B^2) - E^2(B)) + E^2(A) + E^2(B)$$

$$\text{LHS} = \text{RHS} \quad (c) \text{ True}$$

[MCQ]

A machine produces items with there different sizes:

$x = 2$, $x = 4$ and $x = 6$ units, with corresponding probabilities $P(x)$ given below:

x	2	4	6
$P(x)$	0.2	0.5	0.3

What is the standard deviation of the produced sizes?

(a) 1.2

(b) 1.48

(c) 1.80

(d) 1.4

$$\begin{aligned} \text{Var} &= E(x^2) - (E(x))^2 \\ &= 19.6 - (4.2)^2 = ? \\ \text{SD}(\sigma) &= + \sqrt{\text{Var}(x)} = \textcircled{d} \checkmark \end{aligned}$$

$$\begin{aligned} E(x) &= \sum p_i x_i \\ &= 0.2(2) + 0.5(4) + 0.3(6) \end{aligned}$$

$$= 0.4 + 2.0 + 1.8$$

$$= 4.2 = \text{Mean}$$

$$\begin{aligned} E(x^2) &= \sum p_i x_i^2 \\ &= 0.2(2)^2 + 0.5(4)^2 + 0.3(6)^2 \\ &= 0.8 + 8.0 + 10.8 \\ &= 19.6 = 2^{\text{nd}} \text{ Moment} \end{aligned}$$

[NAT]

If a random variable X assumes only positive integral values, with the probability

$$P(X = x) = \frac{2}{3} \left(\frac{1}{3} \right)^{x-1}, x = 1, 2, 3, \dots$$

then $E(X)$ is (a) 2.0 (b) 3.5 (c) 1.0 (d) 1.5

$x:$	1	2	3	4	5	
$P(x):$	$\frac{2}{3}$	$\frac{2}{3} \cdot \frac{1}{3}$	$\frac{2}{3} \left(\frac{1}{3} \right)^2$	$\frac{2}{3} \left(\frac{1}{3} \right)^3$	$\frac{2}{3} \left(\frac{1}{3} \right)^4$	

$$\begin{aligned} E(X) &= \sum p_i X_i = \frac{2}{3} \times 1 + \frac{2}{3} \left(\frac{1}{3} \right) (2) + \frac{2}{3} \left(\frac{1}{3} \right)^2 (3) + \frac{2}{3} \left(\frac{1}{3} \right)^3 (4) + \dots \\ &= \frac{2}{3} \left[1 + 2 \left(\frac{1}{3} \right) + 3 \left(\frac{1}{3} \right)^2 + 4 \left(\frac{1}{3} \right)^3 + \dots \right] = \frac{2}{3} \left[1 - \frac{1}{3} \right]^{-2} = 1.5 \end{aligned} \quad \text{(d)}$$

$$\begin{aligned} 1 + 2x + 3x^2 + 4x^3 + \dots \\ = (1-x)^{-2} \end{aligned}$$

BINOMIAL (with Replacement)

① Bernoulli Trial $\begin{cases} \text{success} = p \\ \text{failure} = q \end{cases}$ & $\boxed{p+q=1}$

② $\boxed{P(X=r) = {}^n C_r p^r q^{n-r}}$

Mean = $\underset{E(X)}{np}$, Var = $\boxed{npq}$, SD = $\sqrt{npq}$

③ N. Conditions:

$n = \text{finite}$, $n = \text{Ind}$, $p = \text{const}$ for each R Exp, R Exp $\begin{cases} \text{Success} \\ \text{failure} \end{cases}$ (Bernoulli Trial)

④

$X = \{ \text{which is Required} \} \rightarrow \text{success}$

Mean > Variance
for B. Dist

⑤ (Binomial Dist) $\Rightarrow$ (There is a feeling of with Replacement)
 $\Leftarrow$

A dice is thrown 8 times, find probability that 3 will show (i) exactly 2 times (ii) At least seven times (iii) at least once

$n = 8$, $X = \{ \text{Number of times 3 occurs} \}$ $\rightarrow$ success
 = finite

$$p = P(3) = \frac{1}{6}, \quad q = P(\bar{3}) = \frac{5}{6}$$

$$P(X = r) = {}^n C_r p^r q^{n-r}$$

$$(1) P(X = 2) = {}^8 C_2 (p)^2 (q)^6$$

$$(2) P(X \geq 7) = P(X = 7) + P(X = 8) \\ = {}^8 C_7 (p)^7 (q)^1 + {}^8 C_8 (p)^8 (q)^0$$

$$(3) P(X \geq 1) = 1 - P(X = 0) = 1 - {}^8 C_0 p^0 q^8 = 1 - \left(\frac{5}{6}\right)^8 \underline{\underline{Ans}}$$

$$\textcircled{*} P(1^{\text{st}} \text{ two throws produces '3'}) = ? = p \times p \times (\text{something occurs})$$

$$= \left(\frac{1}{6}\right)^2 \times (1)^6 = \frac{1}{36}$$

$$\textcircled{*} P(\text{only } 1^{\text{st}} \text{ two throws produces '3'}) = ? = P(\underline{3} \underline{3} \underline{\bar{3}} \underline{\bar{3}} \underline{\bar{3}} \underline{\bar{3}} \underline{\bar{3}} \underline{\bar{3}})$$

$$= p^2 q^6 = ?$$

$$\textcircled{*} P(\text{exactly 2 throws produces '3'}) = ? = {}^8C_2 p^2 q^6$$

only 3rd Part is Based on Binomial Dist.

In an experiment, positive and negative value are equally likely to occur. The probability of obtaining at most one negative value in five trials is

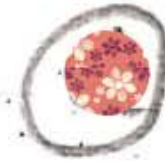
(a) $\frac{1}{32}$

Success

(b) $\frac{2}{32}$

(c) $\frac{3}{32}$

(d) $\frac{6}{32}$



$$P(\text{getting true value}) = \frac{1}{2} = q \quad X = \{ \text{No. of times -ve value is occurring} \}$$

$$P(\text{" -ve "}) = \frac{1}{2} = p \quad n = 5$$

Success

$$P(X \leq 1) = P(X=0) + P(X=1)$$

$$= {}^5C_0 (p)^0 (q)^5 + {}^5C_1 (p)^1 (q)^4 = \frac{1}{32} + \frac{5}{32} = \frac{6}{32}$$

The probability that a screw manufactured by a company is defective is 0.1. The company sells screws in packets containing 5 screws and gives a guarantee of replacement if one or more screws in the packet are found to be defective. The probability that a packet would have to be replaced is _____.



~~A 0.21 B 0.41~~

C 0.90 D None

$X = \{ \text{Number of Defective screws} \}$ Success

$n = 5, p = P(\text{Def}) = 0.1, q = P(\text{Non Def}) = 0.9$

$$P(X \geq 1) = 1 - P(X = 0) = 1 - {}^5C_0 (0.1)^0 (0.9)^5 = 1 - \left(\frac{9}{10}\right)^5 = ? = 0.41$$

[MSQ]

A coin is tossed 5 times. let X denote the number of heads obtained. Assuming the coin is fair, Which of the following statements about X is true?

- (a) ✓ X follows a binomial distribution with parameter $n = 5$ & $p = 0.5$. (T)
- (b) ✓ The mean of X is 2.5 (T)
- (c) The variance of X is 125. (F)
- (d) ✓ The probability of obtaining exactly 3 heads is ${}^5C_3 (0.5)^3 (0.5)^2$ (T)

$n=5$, Head = Success $\rightarrow p = \frac{1}{2}$
 $q = \frac{1}{2}$

$$X \sim B\{n, p\}$$

$$\text{Mean} = np = 5 \times \frac{1}{2} = 2.5 \checkmark$$

$$\text{Var} = npq = 5 \times \frac{1}{2} \times \frac{1}{2} = 1.25$$

$$P(X=3H) = {}^5C_3 (p)^3 (q)^2$$

POISSON

$\lambda \rightarrow$ Average per unit Data,

$$\text{Mean} = \text{Var} = \lambda$$

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

h.l. limit:

$n \rightarrow \infty$ (very large)

$p \rightarrow 0$ (very small)

$np \rightarrow \text{const} (= \lambda)$

$$X \sim P\{\lambda\}$$

✓ (*) n is not sure but we know λ then we should apply P. Dist.

(*) B. Dist $\rightarrow$ P. Dist
 $(n, p) \not\Leftarrow (\lambda)$

(*) $n \rightarrow \infty$
 But $p \rightarrow 0$ } Normal Dist.

Consider a random variable to which a Poisson distribution is best fitted. It happens that

$$P_{(x=1)} = \frac{2}{3} P_{(x=2)}$$
 on this distribution plot. The

variance of this distribution will be

(a) 3

(b) 2

(c) 1

(d) $\frac{2}{3}$



$$P(x=1) = \frac{2}{3} P(x=2)$$

$$\frac{e^{-\lambda} \lambda^1}{1!} = \frac{2}{3} \left(\frac{e^{-\lambda} \lambda^2}{2!} \right)$$

$$3\lambda = \lambda^2$$

$$\lambda(\lambda - 3) = 0$$

$$\lambda = 0 \text{ \& \; } \lambda = 3$$

(avoid)

ie Variance = $\lambda = 3$

(a) ✓

A manufacturer knows that the razor blade he makes contain on an average 0.5% defectives. He pack them in a packet of 5. Then the probability (%) that a packet picked at random will contain 3 or more faulty blades is

(a) 2.55×10^{-6}

(c) 25.5×10^{-4}

~~(b) 2.55×10^{-4}~~

(d) 25.5×10^{-6}

$\lambda = np = 5 \times \frac{5}{1000} = \frac{1}{40}$



$n = 5$

$X = \{ \text{Number of faulty Blades} \}$

Blade $\begin{cases} \text{Faulty Blade} \Rightarrow p = 0.5\% = 0.005 = \frac{5}{1000} \\ \text{Non faulty Blade} \Rightarrow q = 0.995 = \frac{995}{1000} \end{cases}$

Success

M-I Binomial Dist.

M-II Poisson Dist.

$$P(X \geq 3) = P(X=3) + P(X=4) + P(X=5)$$

$$= \text{Do yourself.}$$

$$P(X \geq 3) = e^{-\lambda} \left[\frac{\lambda^3}{3!} + \frac{\lambda^4}{4!} + \frac{\lambda^5}{5!} \right]$$

$$= 2.55 \times 10^{-4}$$

[MCQ]

In a chemical plant, the probability of a minor leakage occurring on any given day is $p = 0.002$. Assuming the events are independent, find the probability that there will be exactly one leakage in a period of 500 days.

- (a) $1/e$ (b) $1/e^2$
(c) $1/e^3$ (d) $1/\sqrt{e}$

M-II $n = 500, p = \frac{1}{500}$
 $\lambda = np = 500 \times \frac{1}{500} = 1$

$$P(X=1) = \frac{e^{-\lambda} \lambda^1}{1!} = e^{-1} = \frac{1}{e}$$

av

$X = \{ \text{Number of leakage per day} \}$ success

$$p = P(\text{leakage per day}) = 0.002 = \frac{2}{1000} = \frac{1}{500} = \frac{1/500}{1}$$

M-I

✓
Average No of Leakage in one day = $\frac{1}{500}$
" " " " in 500 days = $\frac{1}{500} \times 500 = 1 = \lambda$

CONTINUOUS RANDOM VARIABLE →



(1) $f(x) \geq 0$ (2) $\int_{-\infty}^{\infty} f(x) dx = 1$ = Total area under $f(x)$

$$0 \leq f(x) < \infty$$

(3) $F(x) = \int_{-\infty}^x f(x) dx$ (C.D.F.) $\leftarrow f(x) = \frac{d}{dx} F(x)$ (p.d.f), $0 \leq F(x) \leq 1$

(4) $P(a < x < b) = \int_a^b f(x) dx = F(b) - F(a)$

(5) Mean $(x) = E(x) = \int_{-\infty}^{\infty} x f(x) dx$ (M.I) (1) Var $(x) = E(x^2) - E^2(x)$

(6) $E(g(x)) = \int_{-\infty}^{\infty} g(x) f(x) dx$

(8) $\sigma = \sqrt{\text{Var}(x)}$

$X = \text{D.R.V.}$

$X:$ $x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5$

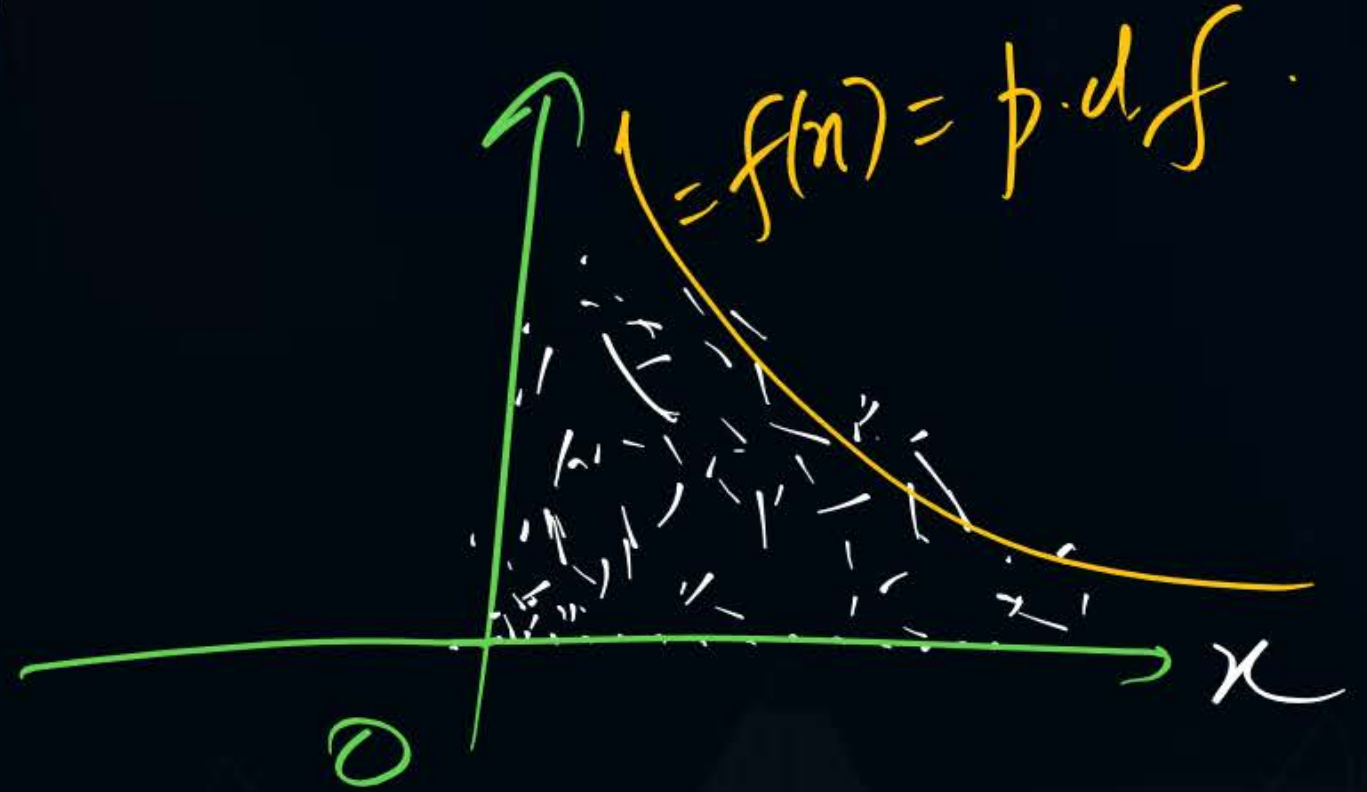
$P(X):$ $p_1 \quad p_2 \quad p_3 \quad p_4 \quad p_5$

$p(x)$

BAR DIAGRAM



$X = \text{C.R.V.}$



If probability density function of a random variable X is $f(x) = x^2$ for $-1 \leq x \leq 1$ and $f(x) = 0$ for any other values of x , then the percentage probability

$P\left(-\frac{1}{3} \leq x \leq \frac{1}{3}\right)$ is CHOORAN QUESTION

PYQ

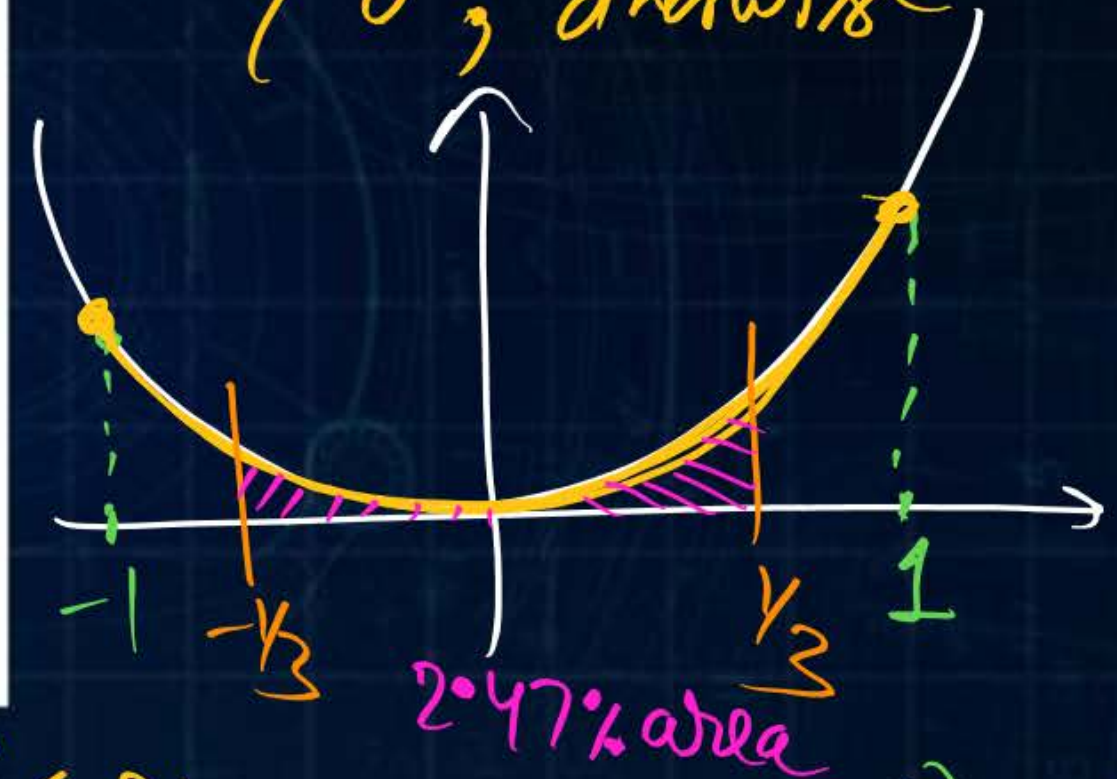
(a) 0.247

(b) 2.47

(c) 24.7

(d) 247

$$f(x) = \begin{cases} x^2, & -1 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$



$$P\left(-\frac{1}{3} \leq x \leq \frac{1}{3}\right) = \int_{-1/3}^{1/3} f(x) dx = 2 \int_0^{1/3} (x^2) dx = 0.0247 \quad (b) \checkmark$$

Even func

% Prob = 0.0247 × 100 = 2.47%

[MCQ]

Consider the function $f(x) = C e^{-x^6}$, where $x \in \mathbb{R}$. For what value of C is function a probability density function?

- (a) $\frac{2}{\Gamma(1/6)}$ (b) $\frac{3}{\Gamma(1/6)}$
(c) $\frac{1}{3\Gamma(1/6)}$ (d) $\frac{3}{2\Gamma(1/6)}$

Total area under $f(x) = 1$.

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\int_{-\infty}^{\infty} C e^{-x^6} dx = 1$$

$$2C \int_0^{\infty} e^{-x^6} dx = 1$$

$$\text{Put } x^6 = y \Rightarrow x = y^{1/6}$$

$$dx = \frac{1}{6} y^{-5/6} dy$$

$$2C \int_0^{\infty} e^{-y} \left(\frac{1}{6} y^{-5/6} dy \right) = 1$$

$$\frac{C}{3} \int_0^{\infty} e^{-y} y^{-1/6} dy = 1$$

$$\frac{C}{3} \Gamma\left(\frac{1}{6}\right) = 1 \Rightarrow C = \frac{3}{\Gamma(1/6)} \text{ (b) } \checkmark$$

[NAT]

Let x be a continuous random variable with P.D.F

$$f_x(x) = \begin{cases} 5x^4; & 0 < x \leq 1 \\ 0 & ; \text{ otherwise} \end{cases}$$

the value of $P(x \leq 2/3 \mid x > 1/3)$ will be: _____

$$\begin{aligned} P\left(x < \frac{2}{3} \mid x > \frac{1}{3}\right) &= \frac{P\left(x < \frac{2}{3} \cap x > \frac{1}{3}\right)}{P\left(x > \frac{1}{3}\right)} \\ &= \frac{P\left(\frac{1}{3} < x < \frac{2}{3}\right)}{P\left(x > \frac{1}{3}\right)} = \frac{\int_{1/3}^{2/3} f(x) dx}{\int_{1/3}^{\infty} f(x) dx} \end{aligned}$$

w.k. Nat $P(A/B) = \frac{P(A \cap B)}{P(B)}$

$$= \frac{\int_{1/3}^{2/3} 5x^4 dx}{\int_{1/3}^1 (5x^4) dx} = ? = \underline{0.128}$$



[MCQ]

Let x be a continuous random variable with density function

$$f(x) = \begin{cases} \frac{x}{9}(4-x); & 0 < x < 3 \\ 0 & ; \text{ otherwise} \end{cases}$$

p.d.f.

What is the mode of x ?

- | | |
|---------|-------|
| (a) 4/9 | (b) 1 |
| (c) 3/2 | (d) 2 |

At Mode (x), $f(x)$ will be max.
ie $f'(x) = 0$ & $f''(x) < 0$

$$f(x) = \frac{4x}{9} - \frac{x^2}{9}, \quad f'(x) = \frac{4}{9} - \frac{2x}{9}$$

$$\text{Put } f'(x) = 0$$

$$\frac{4}{9} - \frac{2x}{9} = 0 \Rightarrow x = 2$$

(d) ✓

EXPONENTIAL DISTRIBUTION

① p.d.f $f(t) = \begin{cases} \mu e^{-\mu t}, & t \geq 0 \\ 0, & t < 0 \end{cases}$

② $\text{Mean}(t) = E(t) = \frac{1}{\mu} = \text{S.D.}$

③ $\mu \rightarrow$ service rate
 $\frac{1}{\mu} \rightarrow$ Av. waiting time

④ C.D.F $F(t) = \begin{cases} 1 - e^{-\mu t}, & t \geq 0 \\ 0, & t < 0 \end{cases}$



④ $P(t_1 < t < t_2) = \int_{t_1}^{t_2} f(t) dt$

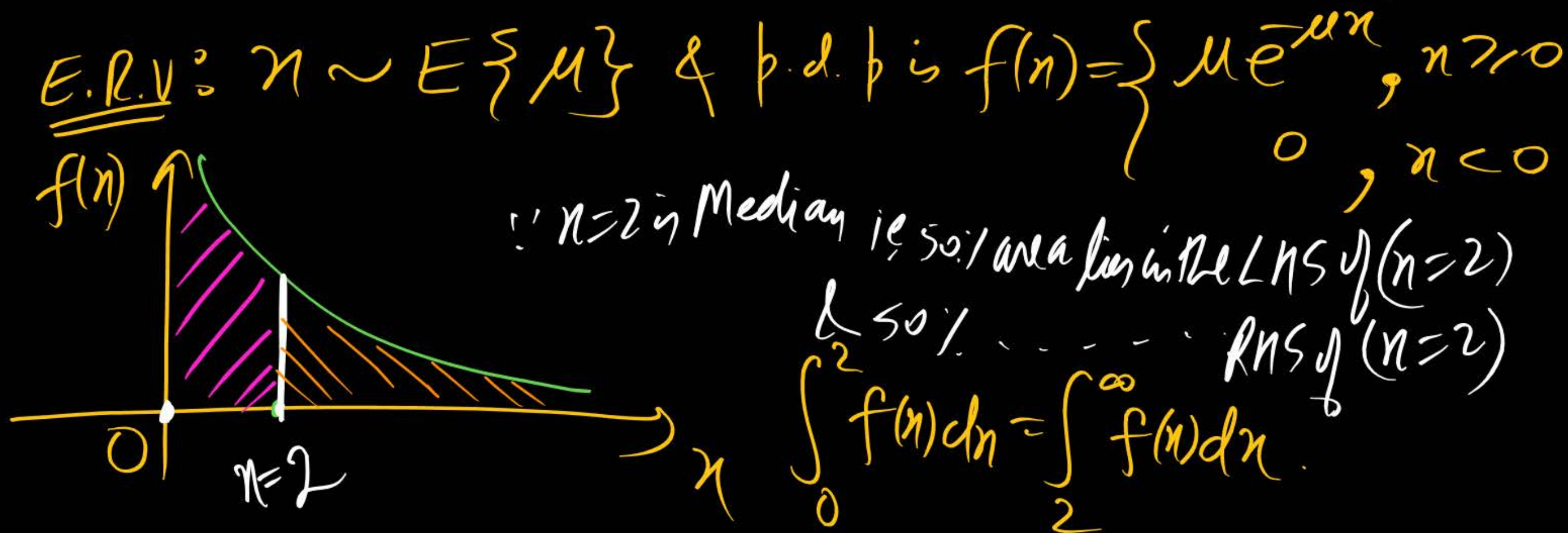
⑤ In P-Dist Mean = Var = $\frac{1}{\mu}$ & S.D. = $\frac{1}{\mu}$
 In Exp Dist Mean = S.D. = $\frac{1}{\mu}$
 & Var = $\frac{1}{\mu^2}$

[NAT]



If 'x' is an exponentially distributed variable ($0 \leq x < \infty$) with median = 2, then the mean of 'x' is ____ (Round off to one decimal place)

$\mu = ?$



$$\int_0^2 \mu e^{-\mu x} dx = \int_2^{\infty} \mu e^{-\mu x} dx$$

.....

$$\boxed{\mu = \frac{1}{2.9}} \text{ so Mean} = \frac{1}{\mu} = 2.9 \text{ } \underline{\underline{Ans.}}$$

The arrival of customers over fixed time intervals in a bank follow a poisson distribution with an average of 30 customers/hour. The probability that the time between successive customer arrival is between 1 and 3 minutes is ____ (correct to two decimal places).

0.38

$$X \sim P\{\lambda = 30 \text{ C/hr}\}$$

$$P(1 < t < 3) = \int_1^3 f(t) dt$$

$$= \dots = 0.38$$

$X = \{ \text{Number of customers in one min} \}$ success

$$\lambda = 30 \text{ Cust/hr} = \frac{30}{60} \text{ Cust/min} = \frac{1}{2} \text{ Cust/min} = 1 \text{ Cust in two min}$$

$$\text{Av. time gaps in two successive Arrivals} = \frac{1}{\mu} = 2 \text{ min} \Rightarrow \mu = \frac{1}{2}$$

$$f(t) = \begin{cases} \mu e^{-\mu t} & t \geq 0 \\ 0 & t < 0 \end{cases}$$

M-II Try to solve it by using Poisson Dist.

Ques I: $P(0 < x < 3) = ?$

Ques II: $P(0 < x < 1) = ?$

Ans $P(1 < x < 3) = \text{Ques I} - \text{Ques II}$
 $= ?? = 0.38$



(Dr Puneet Sirpw)



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Thank
YOU